CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 10: Arithmetic Progression

Topic-wise short notes with solved examples

An arithmetic progression is a sequence in which each term differs from the one before it by a fixed number. This chapter covers its general term, the sum of its terms and the arithmetic mean.

1 Sequence, series and A.P.

- A **sequence** is a list of numbers written in a definite order by a rule. When the terms are joined by plus signs, it is a **series**.
- An **arithmetic progression (A.P.)** is a sequence in which the difference between any term and the term before it is constant.
- The constant is the **common difference** d, and the first term is a.

$$d = t_2 - t_1 = t_3 - t_2 = \dots \qquad \text{General A.P.: } a,\ a+d,\ a+2d,\ a+3d,\ \dots$$

- d can be positive (terms increase), negative (terms decrease) or zero (all terms equal).

Example 1.1 Is the sequence $-8, -5, -2, 1, \dots$ an A.P.? If so, find its common difference and the next term.

SOLUTION

$-5-(-8) = 3, -2-(-5) = 3, 1-(-2) = 3$. The differences are equal.

Yes, it is an A.P. with $d = 3$; the next term is 4.

2 The general term

- The nth term (general term) of an A.P. is

$$t_n = a + (n-1)d$$

- If the last term is l, the nth term from the end is $l - (n-1)d$.
- A sequence is an A.P. exactly when its nth term is a linear expression in n, such as $t_n = 5n - 2$.

Example 2.1 Find the 15th term of the A.P. $6, 10, 14, \dots$

SOLUTION

$a = 6, d = 4$

$t_{15} = 6 + 14 \times 4$

$t_{15} = 62$

Example 2.2 Which term of the A.P. $3, 8, 13, \ldots$ is 148?

SOLUTION

$3 + (n-1) \times 5 = 148$, so $n - 1 = 29$

The 30th term.

Example 2.3 The 6th term of an A.P. is 19 and its 16th term is 59. Find the first term, the common difference and the general term.

SOLUTION

$a + 5d = 19$ and $a + 15d = 59$

Subtract: $10d = 40$, so $d = 4$ and $a = 19 - 20 = -1$

$t_n = -1 + (n-1) \times 4 = 4n - 5$

$a = -1,\ d = 4,\ t_n = 4n - 5$

Example 2.4 The nth term of a sequence is $t_n = 7 - 3n$. Show that it is an A.P. and find its common difference.

SOLUTION

$t_n - t_{n-1} = (7 - 3n) - [\,7 - 3(n-1)\,] = -3$, which is constant.

It is an A.P. with $d = -3$.

3 Sum of n terms

$$S_n = \frac{n}{2}\,[\,2a + (n-1)d\,] \qquad S_n = \frac{n}{2}\,(a + l)$$

- Use the second form when the last term l is known.
- $t_n = S_n - S_{n-1}$.

Example 3.1 Find the sum of the first 25 terms of the A.P. $4, 7, 10, \ldots$

SOLUTION

$a = 4,\ d = 3,\ n = 25$

$S_{25} = \frac{25}{2}\,[\,8 + 24 \times 3\,] = \frac{25}{2} \times 80$

$S_{25} = 1000$

Example 3.2 Find the sum of all two-digit numbers that are divisible by 5.

SOLUTION

The numbers are $10, 15, 20, \ldots, 95$: $a = 10$, $d = 5$, $l = 95$.

$10 + (n-1) \times 5 = 95$ gives $n = 18$

$S_{18} = \frac{18}{2}(10 + 95) = 9 \times 105$

The sum is 945.

Example 3.3 How many terms of the A.P. $27, 24, 21, \ldots$ must be taken so that their sum is 0?

SOLUTION

$\frac{n}{2}[54 + (n-1)(-3)] = 0$, so $n(57 - 3n) = 0$

$n = 0$ is rejected.

$n = 19$ terms.

4 Arithmetic mean

- If a, A, b are in A.P., then A is the **arithmetic mean** of a and b.

$$A = \frac{a+b}{2}$$

- Three numbers a, b, c are in A.P. if and only if $2b = a + c$.

Example 4.1 Find the value of k for which $2k + 1$, 13 and $5k - 3$ are three consecutive terms of an A.P.

SOLUTION

$2 \times 13 = (2k + 1) + (5k - 3)$

$26 = 7k - 2$

$k = 4$

5 Choosing terms conveniently

Number of terms	Take them as
3	$a - d,\ a,\ a + d$
4	$a - 3d,\ a - d,\ a + d,\ a + 3d$
5	$a - 2d,\ a - d,\ a,\ a + d,\ a + 2d$

Example 5.1 The sum of three numbers in A.P. is 24 and their product is 440. Find the numbers.

SOLUTION

Let the numbers be $a-d$, a, $a+d$. Their sum is $3a=24$, so $a=8$.

Product: $8(8-d)(8+d)=440$, so $64-d^2=55$ and $d=\pm 3$

The numbers are 5, 8 and 11.

6 Common mistakes

- Using n in place of $n-1$ in the general term.
- Taking d as (first term $-$ second term).
- Accepting a value of n that is negative, zero or a fraction.
- Confusing t_n (a term) with S_n (a sum).

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